

SINGLE QUADRATIC EQUATIONS DO NOT COMPRESS INTEGER DIMENSION

HIROKI FUKUI

ABSTRACT. Call a relation on $\mathbb{Z}^k$ *single-quadratic* if it is defined by one polynomial equation with integer coefficients and total degree at most two, allowing any finite number of existentially quantified integer unknowns. We show that a single-quadratic function $\mathbb{Z}^n \rightarrow \mathbb{Z}^m$ can be injective only if $n \leq m$, and that every $n \leq m$ is attained. In particular no single-quadratic function codes pairs of integers by integers, although Diophantine relations of higher degree do, and although the Cantor polynomial does so on $\mathbb{N}$. The proof rests on a structural dichotomy: a total function whose graph is single-quadratic either grows at most linearly, or agrees on a coset $\rho + M\mathbb{Z}^n$ ($M \geq 1$) with a vector of rational polynomials of degree at most two. The dichotomy is proved by a case analysis of the quadratic part of the defining equation over $\mathbb{Q}$; the indefinite anisotropic case uses units of a real quadratic order. As further consequences, the graph of an integer polynomial is single-quadratic exactly when its degree is at most two, and the class is closed neither under composition nor under conjunction.

1. INTRODUCTION

1.1. Coding integers by integers. Coding a pair of numbers by one number is the basic step of arithmetization. Over the natural numbers $\mathbb{N} = \{0, 1, 2, \dots\}$ a polynomial does it: the Cantor pairing $(x, y) \mapsto \binom{x+y+1}{2} + x$ is a bijection $\mathbb{N}^2 \rightarrow \mathbb{N}$ of degree two, and by the Fueter–Pólya theorem [FP23; see Vse02, Nat16 for modern proofs] the two Cantor polynomials are the only quadratic polynomials with this property.

Over the integers the question of a bijection $\mathbb{Z}^2 \rightarrow \mathbb{Z}$ given by a polynomial with integer coefficients was posed by Hammer as Problem 6028 [Ham75], in connection with pairing functions in recursion theory [Lew81]; Lew [Lew81, Problem B] recorded it as unsolved, next to the corresponding problem over $\mathbb{N}$. For injections, degree is the dividing line. Lew and Rosenberg [LR78a, Theorem 5.3] showed that no quadratic polynomial maps $\mathbb{Z}^2$ injectively into $\mathbb{N}$, and that a quartic one does. Their proof of the first statement uses only that the values on $\mathbb{Z}^2$ are integers, so it applies equally to injective maps into $\mathbb{Z}$ [LR78a, pp. 211–212]. Their quartic is a composition: a quadratic storing function of $\mathbb{N}^2$ of Cantor type applied to two copies of an injective quadratic map $\mathbb{Z} \rightarrow \mathbb{N}$ [LR78a, Theorems 2.5, 4.5, 5.3(3)]. For instance, composing the Cantor bijection with $x \mapsto 2x^2 + x$ gives the injective

2020 *Mathematics Subject Classification.* Primary 11D09; Secondary 11U05, 03D25, 68V15.

Key words and phrases. Diophantine definability; quadratic Diophantine equations; existential definability; injective coding; Pell equation; formal verification.

ORCID 0009-0008-7122-522X.

polynomial $(x, y) \mapsto C(2x^2 + x, 2y^2 + y)$ of degree four on $\mathbb{Z}^2$. Poonen later made the same observation (reported in [Cor99, p. 6], with $2x^2 + x + 1$ and the Cantor bijection of $\mathbb{Z}_{>0}^2$). The map $x \mapsto 2x^2 + x$ takes $0, -1, 1, -2, 2, \dots$ to $0, 1, 3, 6, 10, \dots$

The resulting equation $2b - 2C(2x^2 + x, 2y^2 + y) = 0$ defines the graph of an injection $\mathbb{Z}^2 \rightarrow \mathbb{Z}$ by one equation of degree four, without unknowns.

Cornelissen [Cor99] placed such codings in a logical setting. In his terms a structure *admits Diophantine storing* if its square embeds into it by an injection under which the square, with its induced relations, is Diophantine [Cor99, Definitions 4]. Over $\mathbb{Z}$ his storing maps are given piecewise by polynomials on finitely many pieces, so their graphs are Diophantine: definable by polynomial equations with existentially quantified integer unknowns. Combining his construction principle with a cover of $\mathbb{Z}^2$ by four quadrant pieces, whose images are Diophantine by Lagrange's four-square theorem, he obtained even a bijection $\mathbb{Z}^2 \rightarrow \mathbb{Z}$ with Diophantine graph [Cor99, p. 6, Proposition 7]. More generally, by the theorem of Davis, Putnam, Robinson and Matiyasevich [DPR61, Mat70] every listable relation is Diophantine (over $\mathbb{Z}$ as well as over $\mathbb{N}$, since $\mathbb{N}$ is Diophantine in $\mathbb{Z}$ by Lagrange's theorem), and by Skolem's reduction [Sko38] one equation of total degree four, with unknowns, suffices for each of them.

This paper asks what remains in degree two once unknowns are allowed. Write SE_2 (single equation, degree two) for the class of relations on integer tuples defined by one polynomial equation of total degree at most two with integer coefficients and any finite number of existentially quantified integer unknowns. Whether a single quadratic equation has an integer solution is decidable [Sie72, GS04], so SE_2 is a decidable fragment of the Diophantine relations. Unknowns enlarge the class: $\mathbb{N} \subseteq \mathbb{Z}$ is in SE_2 , via $a = u_1^2 + u_2^2 + u_3^2 + u_4^2$, although a quadratic equation in a alone defines at most two points or all of $\mathbb{Z}$. Even without unknowns the class contains functions that are not polynomials, such as $a \mapsto \lceil a/2 \rceil$ below. On the other hand SE_2 is closed neither under composition nor under conjunction (Corollary 4), the closure properties on which the usual reduction of arity by iterated pairing relies. We show that none of the extra room helps: one quadratic equation cannot compress dimension, with any number of unknowns.

Over $\mathbb{Q}$ the corresponding question for polynomials has a separate history: in reply to a question of H. Friedman, Zagier suggested $x^7 + 3y^7$, and Cornelissen noted that a generalized *abc*-conjecture for $\mathbb{Q}$ implies that $x^N + 3y^N$ is injective on $\mathbb{Q}^2$ for all large odd N [Cor99, Remarque 10]. Poonen [Poo10] derived the existence of a polynomial injection $\mathbb{Q} \times \mathbb{Q} \rightarrow \mathbb{Q}$ from a hypothesis implied by the Bombieri–Lang conjecture, and Pasten [Pas20] proved unconditionally that, over any number field, suitable affine surfaces built from an elliptic curve of positive rank carry a regular function injective on their rational points. Those results concern fields; ours concern degree two over $\mathbb{Z}$, with unknowns.

1.2. Results. Conventions. $\mathbb{N} = \{0, 1, 2, \dots\}$. For $a \in \mathbb{Z}^n$, $\|a\|_\infty = \max_i |a_i|$ (and $\|a\|_\infty = 0$ for $n = 0$). Degrees are total degrees; the zero polynomial has degree at most two.

A total function $f : \mathbb{Z}^n \rightarrow \mathbb{Z}^m$ is *SE₂-definable* if its graph $\{(a, f(a))\} \subseteq \mathbb{Z}^{n+m}$ is in SE_2 : there are $r \geq 0$ and $P \in \mathbb{Z}[a_1, \dots, a_n, b_1, \dots, b_m, u_1, \dots, u_r]$ of total degree at most two such that for all $a \in \mathbb{Z}^n$, $b \in \mathbb{Z}^m$,

$$b = f(a) \iff \exists u \in \mathbb{Z}^r \ P(a, b, u) = 0.$$

Theorem 1. *Let $n, m \geq 0$. An injective SE_2 -definable function $\mathbb{Z}^n \rightarrow \mathbb{Z}^m$ exists if and only if $n \leq m$.*

The *coding arity* of SE_2 over $\mathbb{Z}$ is the largest n for which some SE_2 -definable function $\mathbb{Z}^n \rightarrow \mathbb{Z}$ is injective. The case $m = 1$ of Theorem 1 says that it is one: no SE_2 -definable function $\mathbb{Z}^2 \rightarrow \mathbb{Z}$ is injective. For Diophantine relations every n is attained (iterate the quartic of [LR78a, Theorem 5.3(3)]), so there is no largest n . If the defining equivalence is required only for inputs $a \in \mathbb{N}^n$, the coding arity of SE_2 is at least two, by the Cantor polynomial (§5).

The sufficiency in Theorem 1 is zero-padding (§5). The necessity follows from a structure theorem.

Theorem 2 (structural dichotomy). *Let $f : \mathbb{Z}^n \rightarrow \mathbb{Z}^m$ be SE_2 -definable. Then at least one of the following holds.*

1. (linear growth) *There is C with $\|f(a)\|_\infty \leq C(1 + \|a\|_\infty)$ for all $a \in \mathbb{Z}^n$.*
2. (coset polynomial form) *There are $M \geq 1$, $\rho \in \mathbb{Z}^n$ and polynomials $g_1, \dots, g_m \in \mathbb{Q}[t_1, \dots, t_n]$ of total degree at most two with $f(\rho + Mt)_j = g_j(t)$ for all $t \in \mathbb{Z}^n$ and all j .*

Theorem 2 gives Theorem 1 through two short lemmas (§3): a counting argument in case 1, and a collision $f(\rho + Mv) = f(\rho - Mv)$ in case 2. Counting alone is not enough: $a \mapsto a^2$ is SE_2 -definable, and a box count that uses only a quadratic growth bound gives $n \leq 2m$. The point of Theorem 2 is that growth faster than linear happens only polynomially on a coset, where the collision is available.

Alternative 1 alone is not enough: $a \mapsto a^2$ (defined by $b - a^2 = 0$) does not grow linearly. Not every SE_2 -definable function is a polynomial. For example, $a \mapsto \lceil a/2 \rceil$ is defined by

$$(2b - a)(2b - a - 1) = 0;$$

it is affine on each residue class modulo 2, but not polynomial on all of $\mathbb{Z}$.

Theorem 2 also decides which polynomial functions are in the class, and from this the failure of closure follows (§6).

Corollary 3. *Let $p \in \mathbb{Z}[x_1, \dots, x_n]$. The function $\mathbb{Z}^n \rightarrow \mathbb{Z}$, $a \mapsto p(a)$, is SE_2 -definable if and only if $\deg p \leq 2$.*

Corollary 4. (a) *The function $a \mapsto a^2$ on $\mathbb{Z}$ is SE_2 -definable and its composition with itself is not.* (b) *The relations $\{(x, y, z) \in \mathbb{Z}^3 : y = x^2\}$ and $\{(x, y, z) \in \mathbb{Z}^3 : z = y^2\}$ are in SE_2 and their intersection is not.*

Without unknowns Corollary 3 is immediate. We have not found the version with unknowns in the literature we searched; its proof here uses only Theorem 2.

1.3. What is new. For $(n, m) = (2, 1)$, no unknowns, and f itself a polynomial, Theorem 1 contains the theorem of Lew and Rosenberg quoted above, in the form given by their proof (injective maps into $\mathbb{Z}$): a quadratic polynomial with real coefficients that takes integer values on $\mathbb{Z}^2$ has twice each coefficient an integer [LR78a, p. 212], so its graph is defined by one quadratic equation after clearing denominators; Cornelissen [Cor99, p. 6] restates it as the non-existence of quadratic storing functions for $\mathbb{Z}$. For quadratic polynomials restricted to the lattice points of a cone, see Gjaldabæk [Gj21]. The present setting differs in two ways: unknowns are allowed, and the defined function need not be a polynomial (the example $\lceil a/2 \rceil$ above). Integer coding by a single equation is possible in degree four even without

unknowns [LR78a, Theorem 5.3(3)]; what is new is that in degree two it is impossible, in every dimension, however many unknowns are used. We have not examined degree three. Most of the work in this paper concerns the two differences.

1.4. Formal verification and use of AI tools. Theorem 1 and the ingredients of the proof of Theorem 2 listed in Appendix A are formalized in the Lean 4 proof assistant with the mathlib library (toolchain `leanprover/lean4:v4.31.0-rc1`, mathlib commit `d46bd45`). The archived development [Fuk-L] contains formal proofs of Theorem 1 and the two statements used to derive Theorem 2; the archived proof of Theorem 1 goes through both. A supplementary development [Fuk-S], built against [Fuk-L] with the same toolchain and mathlib revision, contains a composed statement of Theorem 2 and formal proofs of Corollaries 3 and 4. The extension of Corollary 3 to integer-valued polynomials with rational coefficients (§6) is proved in this paper only. For each row of Appendix A the statement in this paper was compared with the formal statement by reading both: the proof checker certifies that the formal statements are proved, and whether they match the statements here rests on that reading. Some formal proofs use constructions different from those given here; the table notes where. Each listed declaration depends only on Lean’s three standard axioms (`propext`, `Classical.choice`, `Quot.sound`); in particular none depends on `sorryAx` or on `Lean.ofReduceBool`. The paper is self-contained and does not depend on the formalization.

Use of AI tools. Large language models were used in this work: Claude and Claude Code (Anthropic) and ChatGPT (OpenAI). Their roles were to explore and compare proof strategies; to search for tactic-level proofs and draft code for the Lean formalization, against formal statements fixed by the author; to draft and revise the text; to read statements, proofs and drafts adversarially; and to assist the literature search. No model output was accepted as a mathematical fact: the formal declarations listed in Appendix A were checked by the Lean kernel, and every definition, statement, proof and citation in this paper was checked by the author, who is responsible for the entire content.

2. THE SHAPE OF THE EQUATION

Fix $P(a, b, u)$ of total degree at most two defining f . Write $z = (b, u) \in \mathbb{Z}^{m+r}$ and $V = \mathbb{Q}^{m+r}$. Because P has degree at most two, it splits uniquely as

$$(2.1) \quad P(a, z) = Q(z) + \ell_a(z) + q(a),$$

where Q is an integral quadratic form in z alone, ℓ_a is linear in z with coefficients affine in a , and q has degree at most two in a . Write

$$B(v, w) = Q(v + w) - Q(v) - Q(w)$$

for the polar form; thus $B(v, v) = 2Q(v)$. The *radical* is $\text{rad } Q = \{e \in V : B(e, \cdot) = 0\}$; a vector x is *isotropic* if $Q(x) = 0$ (we allow $x = 0$), so radical vectors are isotropic. Let $\pi : V \rightarrow \mathbb{Q}^m$ be the projection onto the output coordinates b .

An integral z with $P(a, z) = 0$ is a *solution at a* . That P defines f means: for each a there is a solution at a (totality), and every integer solution has $\pi(z) = f(a)$ (output uniqueness). Nothing is assumed about the unknowns; two solutions with the same output and different unknowns are allowed.

Lemma 2.1 (polar image). *The image of the map $V \rightarrow V^*$, $x \mapsto B(x, \cdot)$, is the annihilator of $\text{rad } Q$.*

Proof. The image is contained in the annihilator by symmetry of B . The kernel of the map is $\text{rad } Q$, so the image has dimension $\dim V - \dim \text{rad } Q$, which is the dimension of the annihilator. $\square$

Lemma 2.2 (the radical is invisible). *Suppose $m \geq 1$, $Q \neq 0$, and P defines f . Then $\ell_a(e) = 0$ and $\pi(e) = 0$ for every $e \in \text{rad } Q$ and every $a \in \mathbb{Q}^n$.*

Proof. Both statements are linear in e , and $\ell_a(e)$ is affine in a ; so it suffices to treat integral e and integral a . For e radical, $Q(z + te) = Q(z) + tB(z, e) + t^2Q(e) = Q(z)$, hence

$$(2.2) \quad P(a, z + te) = P(a, z) + t\ell_a(e).$$

Let z_0 be a solution at a , and suppose $k = \ell_a(e) \neq 0$. For $y \in \mathbb{Z}^{m+r}$,

$$P(a, z_0 + ky) = k(B(z_0, y) + \ell_a(y) + kQ(y)),$$

so by (2.2) the integral vector $z_y = z_0 + ky + t_y e$ with $t_y = -(B(z_0, y) + \ell_a(y) + kQ(y))$ is a solution at a . Output uniqueness gives $k\pi(y) + t_y\pi(e) = 0$ for all y . If $\pi(e) = 0$ this says $\pi(y) = 0$ for all integral y , impossible since $m \geq 1$. If $\pi(e) \neq 0$, replace y by sy ($s \in \mathbb{Z}$): the identity $ks\pi(y) = (s(B(z_0, y) + \ell_a(y)) + ks^2Q(y))\pi(e)$ for all s forces $kQ(y)\pi(e) = 0$, i.e. $Q(y) = 0$ for every integral y , contradicting $Q \neq 0$. Hence $\ell_a(e) = 0$. Then $z_0 + e$ is a solution by (2.2), and output uniqueness gives $\pi(e) = 0$. $\square$

Corollary 2.3 (affine centre). *If $m \geq 1$, $Q \neq 0$ and P defines f , there are $c_0 \in V$ and $c_1, \dots, c_n \in V$ such that $c(a) = c_0 + \sum a_i c_i$ satisfies $B(c(a), \cdot) = -\ell_a$ for every a . With $R(a) = Q(c(a)) - q(a)$, a polynomial of degree at most two,*

$$(2.3) \quad P(a, z) = 0 \iff Q(z - c(a)) = R(a).$$

Proof. Write $\ell_a = \lambda_0 + \sum a_i \lambda_i$. By Lemma 2.2 each λ_i annihilates the radical, so by Lemma 2.1 it is $-B(c_i, \cdot)$ for some c_i . Then $Q(z - c) = Q(z) - B(z, c) + Q(c) = Q(z) + \ell_a(z) + Q(c)$, which gives (2.3). $\square$

We call $z - c(a)$ the *off-centre part* of a solution z at a .

3. FROM THE DICHOTOMY TO THE DIMENSION BOUND

Lemma 3.1. *If $f : \mathbb{Z}^n \rightarrow \mathbb{Z}^m$ is injective and has linear growth, then $n \leq m$.*

Proof. Let $\|f(a)\|_\infty \leq C(1 + \|a\|_\infty)$ with $C \geq 1$ an integer, and suppose $n \geq m + 1$. The box $\|a\|_\infty \leq N$ has $(2N + 1)^n$ points, and f maps it injectively into the box of radius $C(1 + N)$, which has $(2C(1 + N) + 1)^m \leq ((2C + 1)(N + 1))^m$ points. Take $N = (2C + 1)^m$. Then

$$(2N + 1)^n \geq (2N + 1)(2N + 1)^m > N(N + 1)^m = ((2C + 1)(N + 1))^m,$$

a contradiction. $\square$

Lemma 3.2. *If f has a coset polynomial form and $m < n$, then f is not injective.*

Proof. Write $g_j = \gamma_j + L_j + H_j$ with L_j linear and H_j homogeneous quadratic. The linear map $L = (L_1, \dots, L_m) : \mathbb{Q}^n \rightarrow \mathbb{Q}^m$ has a nonzero kernel since $m < n$; clearing denominators gives $v \in \mathbb{Z}^n \setminus \{0\}$ with $L(v) = 0$. Since $H_j(-v) = H_j(v)$,

$$g_j(v) - g_j(-v) = 2L_j(v) = 0 \quad \text{for all } j,$$

so $f(\rho + Mv) = f(\rho - Mv)$, while $\rho + Mv \neq \rho - Mv$. $\square$

For $(n, m) = (2, 1)$ and polynomial f , Lew and Rosenberg also argue by explicit collisions: $f(x, y) = f(-x, -y)$ when f has no linear terms, and an explicit one-parameter family of coinciding pairs otherwise [LR78a, pp. 211–212]. No discriminant condition and no count enters. Theorem 2 and Lemmas 3.1 and 3.2 together give the necessity in Theorem 1.

4. PROOF OF THEOREM 2

If $m = 0$, alternative 2 holds with the empty family. If $n = 0$, f is constant and alternative 2 holds with $M = 1$. Assume $n, m \geq 1$ from now on. The case $Q \equiv 0$ is §4.1; in §§4.2–4.5 we assume $Q \not\equiv 0$, so that Lemma 2.2 and Corollary 2.3 apply.

The branch split. Over $\mathbb{Q}$, at least one of the following holds:

1. $Q \equiv 0$;
2. some $x \in V$ is isotropic and not radical;
3. every isotropic vector is radical, and (a) $Q \geq 0$ on V , or (b) $Q \leq 0$ on V , or (c) Q takes both signs on V .

The cases cover all possibilities; they need not be disjoint. The conditions concern the form Q , not a coefficient matrix of P (which need not be symmetric).

4.1. The case $Q \equiv 0$. The equation is linear in z : $\sum_j \beta_j(a) b_j + \sum_k \delta_k(a) u_k + q(a) = 0$ with β_j, δ_k affine.

The unknowns drop out. Suppose $\delta_k(a) \neq 0$ for some integer a and some k . Take a solution z_0 at a and an output coordinate j , and put $e = \delta_k(a) e_{b_j} - \beta_j(a) e_{u_k}$. The linear part vanishes on e , so $z_0 + e$ is a solution, and its j -th output differs from that of z_0 by $\delta_k(a) \neq 0$. This contradicts output uniqueness. (A vector moving only the unknowns would prove nothing.) Hence every δ_k vanishes on $\mathbb{Z}^n$, hence identically.

At most one output. Now the equation is $\beta(a) \cdot b + q(a) = 0$ with $\beta(a) \in \mathbb{Z}^m$. If $\beta(a) = 0$ at some a , every b or no b is a solution, contradicting totality and uniqueness. If $m \geq 2$, a nonzero integer k with $\beta(a) \cdot k = 0$ moves a solution to another one. So $m = 1$, and $f(a) = -q(a)/\beta(a)$ with $\beta(a) \neq 0$ and $\beta(a) \mid q(a)$ for all integer a .

The divisor. Write $\beta(a) = \beta^{(1)} \cdot a + d$. If $\beta^{(1)} = 0$, then $f = -q/d$ is a rational polynomial of degree at most two on $\mathbb{Z}^n$. Otherwise pick i with $p = \beta_i^{(1)} \neq 0$, put $t = a_i$ and let x be the other inputs, so that $\beta = pt + h_0(x)$ and $q = At^2 + q_1(x)t + q_0(x)$ with A constant and h_0, q_1 affine. Division gives the polynomial identity

$$p^2 q = (pAt + pq_1 - Ah_0) \beta + [p^2 q_0 - pq_1 h_0 + Ah_0^2],$$

whose bracket does not involve t . The cofactor $pAt + pq_1 - Ah_0$ takes integer values at integral (t, x) , since p, A and the coefficients of q_1 and h_0 are integers. For fixed x , $\beta(t, x)$ divides $q(t, x)$ for every integer t , so it divides the bracket; as $|\beta(t, x)| \rightarrow \infty$ with t , the bracket is 0. Hence $f = -(pAt + pq_1 - Ah_0)/p^2$ is a rational affine function. (The divisor β need not be constant: $(2a + 1)(b - a) = 0$ is an example.)

In this case alternative 2 holds with $M = 1$.

4.2. The case of an isotropic non-radical vector. Let $c(a)$ and R be as in Corollary 2.3.

(a) *Every off-centre part is radical.* If for every integer a every solution z has $z - c(a) \in \text{rad } Q$, then by Lemma 2.2, $f(a) = \pi(z) = \pi(c(a))$, an affine function of a . Alternative 1 holds.

(b) *Some off-centre part is not radical.* Let z_0 be a solution at a_0 with $v_0 = z_0 - c(a_0) \notin \text{rad } Q$; thus $Q(v_0) = R(a_0)$.

Step 1: an isotropic direction seen by v_0 . We find e with $Q(e) = 0$ and $B(v_0, e) \neq 0$. Let x be isotropic and not radical. If $B(v_0, x) \neq 0$, take $e = x$. Otherwise choose $w \in V$ outside the kernels of the two nonzero functionals $B(x, \cdot)$ and $B(v_0, \cdot)$, and put

$$e = Q(w)x - B(x, w)w.$$

Then $Q(e) = Q(w)^2Q(x) - Q(w)B(x, w)^2 + B(x, w)^2Q(w) = 0$ and $B(v_0, e) = -B(x, w)B(v_0, w) \neq 0$.

Step 2: a rational section. Since $Q(e) = 0$, $Q(v_0 + te) = Q(v_0) + tB(v_0, e)$. Put

$$s(a) = c(a) + v_0 + \frac{R(a) - R(a_0)}{B(v_0, e)} e.$$

Then $Q(s(a) - c(a)) = R(a)$, so $s(a)$ is a rational solution at every input a , and each coordinate of s is a rational polynomial of degree at most two in a , with $s(a_0) = z_0$.

Step 3: integrality on a coset. Write $s(a_0 + Mt) = z_0 + M S_1(t) + M^2 S_2(t)$, where S_1, S_2 have rational coefficients independent of M and S_1 is linear, S_2 homogeneous quadratic in t . Choosing $M \geq 1$ divisible by all denominators of S_1 and S_2 makes $s(a_0 + Mt)$ integral for every $t \in \mathbb{Z}^n$.

Step 4: the section is the output. $s(a_0 + Mt)$ is then an integer solution at $a_0 + Mt$, so by output uniqueness $f(a_0 + Mt) = \pi(s(a_0 + Mt))$, a vector of rational polynomials of degree at most two in t . Alternative 2 holds with $\rho = a_0$.

4.3. The case $Q \geq 0$ with every isotropic vector radical. (The hypothesis that every isotropic vector is radical is not used in this case.) By Lemma 2.2 each output coordinate π_j annihilates $\text{rad } Q$, so by Lemma 2.1 there is $w_j \in V$ with $B(w_j, \cdot) = \pi_j$. For a positive semidefinite form, $t^2Q(v) + tB(v, w) + Q(w) = Q(tv + w) \geq 0$ for all rational t gives

$$(4.1) \quad B(v, w)^2 \leq 4Q(v)Q(w).$$

(If $Q(v) = 0$ the left side is a nonnegative linear function of t , so $B(v, w) = 0$. The constant is 4 because B is the full polar form. With constant 1 the inequality fails: $Q(x) = x^2$, $v = 1$, $w = 1/2$.) Let z be a solution at a and $v = z - c(a)$; then $Q(v) = R(a) \geq 0$, and

$$|f(a)_j| = |\pi_j(c(a)) + B(w_j, v)| \leq |\pi_j(c(a))| + 2\sqrt{Q(w_j)}\sqrt{R(a)}.$$

Since c is affine and R has degree at most two, $R(a) \leq C_R(1 + \|a\|_\infty)^2$ and $|\pi_j(c(a))| \leq C_j(1 + \|a\|_\infty)$, so the right-hand side is at most a constant times $1 + \|a\|_\infty$. Alternative 1 holds.

4.4. The case $Q \leq 0$ with every isotropic vector radical. Replace P by $-P$. This defines the same function and replaces Q by $-Q \geq 0$; apply §4.3.

4.5. The case where Q takes both signs and every isotropic vector is radical. We show that every off-centre part is radical; then $f(a) = \pi(c(a))$ is affine as in §4.2(a), and alternative 1 holds.

Suppose instead that a solution z_0 at a_0 has $v = z_0 - c(a_0) \notin \text{rad } Q$. Then $Q(v) \neq 0$, since every isotropic vector is radical.

A visible orthogonal direction of the opposite sign. Let $v^\perp = \{y : B(v, y) = 0\}$, so $V = \mathbb{Q}v \oplus v^\perp$ and $Q(\alpha v + y) = \alpha^2Q(v) + Q(y)$. If Q restricted to $v^\perp$ never had

the sign opposite to $Q(v)$, then Q would not take both signs. So some $y_0 \in v^\perp$ has $Q(y_0)Q(v) < 0$. If $\pi(v) \neq 0$ or $\pi(y_0) \neq 0$, put $w = y_0$. Otherwise $\pi(v) = \pi(y_0) = 0$; then π does not vanish on $v^\perp$ (it is nonzero on $V = \mathbb{Q}v \oplus v^\perp$, as $m \geq 1$), and for $u \in v^\perp$ with $\pi(u) \neq 0$ and a small nonzero rational ε , $w = y_0 + \varepsilon u$ has $Q(w)Q(v) < 0$ and $\pi(w) = \varepsilon\pi(u) \neq 0$. In both cases $B(v, w) = 0$, $Q(v)Q(w) < 0$, and $\pi(v) \neq 0$ or $\pi(w) \neq 0$.

The discriminant is not a square. If $-Q(v)Q(w) = s^2$ with $s \in \mathbb{Q}$, then $s \neq 0$, and $e = sv + Q(v)w$ satisfies $Q(e) = Q(v)(s^2 + Q(v)Q(w)) = 0$ and $B(e, v) = 2sQ(v) \neq 0$: an isotropic vector that is not radical, which this case excludes.

A Pell unit. Let $h \geq 1$ make $A = hQ(v)$ and $A' = hQ(w)$ integers, and put $\Delta = -AA' > 0$, not a square. Let $N \geq 1$ clear the denominators of v and w . Since Δ is a positive non-square, the equation $X^2 - \Delta Y^2 = 1$ has an integer solution with $Y \neq 0$ (Lagrange; see [NZM91, §7.8]); choose one with $X, Y > 0$. The powers of $\eta = X + Y\sqrt{\Delta} > 1$ are pairwise distinct units of $\mathbb{Z}[\sqrt{\Delta}]$; two of them agree modulo $N\mathbb{Z}[\sqrt{\Delta}]$, and since η is a unit, a positive power $\eta^k = X + Y\sqrt{\Delta}$ satisfies $X \equiv 1$, $Y \equiv 0 \pmod{N}$, with $X > 1$ and $Y \neq 0$.

Three solutions at one input. Put

$$z_\pm = c(a_0) + Xv \pm AYw = z_0 + (X - 1)v \pm AYw.$$

These are integral by the congruences. Since $B(v, w) = 0$ and $A^2Q(w) = -\Delta Q(v)$,

$$Q(Xv \pm AYw) = X^2Q(v) + A^2Y^2Q(w) = Q(v)(X^2 - \Delta Y^2) = Q(v) = R(a_0),$$

so z_+ , z_- , z_0 are solutions at a_0 . If $\pi(w) \neq 0$, then $\pi(z_+) - \pi(z_-) = 2AY\pi(w) \neq 0$. If $\pi(w) = 0$, then $\pi(v) \neq 0$ and $\pi(z_+) - \pi(z_0) = (X - 1)\pi(v) \neq 0$. Either way output uniqueness fails. The input a_0 is fixed throughout; only the output and the unknowns move.

4.6. Conclusion. The cases of the branch split are covered by §§4.1–4.5, and each yields alternative 1 or 2. This proves Theorem 2. $\square$

5. SHARPNESS AND THE CODING ARITY

For $n \leq m$ the map $\text{pad}(a) = (a_1, \dots, a_n, 0, \dots, 0)$ is injective, and its graph is defined without unknowns by

$$\sum_{j=1}^m (b_j - \text{pad}(a)_j)^2 = 0.$$

This proves the sufficiency in Theorem 1, including the boundary cases $n = 0$ and $m = 0$. With $m = 1$ it shows that the coding arity of SE_2 is at least one; Theorem 1 shows that it is exactly one.

Over natural-number inputs the answer is different: the Cantor polynomial is a degree-two injection $\mathbb{N}^2 \rightarrow \mathbb{N}$ with graph defined by one equation. The Cantor polynomial satisfies alternative 2 of Theorem 2 on $\mathbb{N}^2$; what fails over $\mathbb{N}$ is Lemma 3.2, whose collision uses the reflected point $\rho - Mv$, which leaves $\mathbb{N}^n$.

6. POLYNOMIAL GRAPHS AND THE FAILURE OF CLOSURE

Proof of Corollary 3. If $\deg p \leq 2$, the graph is defined by $b - p(a) = 0$, without unknowns. Conversely, let $a \mapsto p(a)$ be SE_2 -definable and suppose $d = \deg p \geq 3$; then $n \geq 1$, since a polynomial in no variables is a constant. Apply Theorem 2 with $m = 1$.

Alternative 1. Let $p_d \neq 0$ be the homogeneous part of degree d of p . A nonzero polynomial with integer coefficients does not vanish on all of $\mathbb{Z}^n$, so there is $v \in \mathbb{Z}^n$ with $p_d(v) \neq 0$. Then $h(t) = p(tv)$ is a polynomial in one variable of degree d with leading coefficient $p_d(v)$, so $|h(t)|/|t|^d \rightarrow |p_d(v)| \geq 1$ as $|t| \rightarrow \infty$. Linear growth gives $|h(t)| \leq C(1 + |t| \|v\|_\infty)$ for all $t \in \mathbb{Z}$. Since $d \geq 2$, the two bounds are incompatible for large $|t|$.

Alternative 2. Let $p(\rho + Mt) = g(t)$ for all $t \in \mathbb{Z}^n$, with $M \geq 1$, $\rho \in \mathbb{Z}^n$ and $g \in \mathbb{Q}[t]$ of degree at most two. The polynomial $G(t) = p(\rho + Mt) - g(t) \in \mathbb{Q}[t]$ vanishes on $\mathbb{Z}^n$, hence is the zero polynomial (induction on n : a nonzero rational polynomial in one variable has finitely many roots). Substitute $t = (x - \rho)/M$; this is a ring homomorphism $\mathbb{Q}[t] \rightarrow \mathbb{Q}[x]$, and it sends $p(\rho + Mt)$ to $p(x)$. Hence $p(x) = g((x - \rho)/M)$ in $\mathbb{Q}[x]$. An affine substitution does not raise the total degree, so $\deg p \leq \deg g \leq 2$, a contradiction. $\square$

The degree bound is the same over $\mathbb{Z}$ and over $\mathbb{Q}$, so it does not matter in which ring the degree of p is read. The statement and the proof extend to integer-valued $p \in \mathbb{Q}[x_1, \dots, x_n]$: clear denominators in $b - p(a) = 0$, and use $|p_d(v)| > 0$ in place of $|p_d(v)| \geq 1$.

For relations, $R \subseteq \mathbb{Z}^k$ is in SE_2 if there are $r \geq 0$ and $P \in \mathbb{Z}[v_1, \dots, v_k, u_1, \dots, u_r]$ of total degree at most two with $v \in R \iff \exists u \in \mathbb{Z}^r P(v, u) = 0$. The coordinates v are free, the u are existentially quantified. The class is closed under existential quantification of a free coordinate: the coordinate becomes one more unknown. The two closure properties that fail are those of Corollary 4.

Proof of Corollary 4. (a) The function $a \mapsto a^2$ is defined by $b - a^2 = 0$. Its composition with itself is $a \mapsto a^4$, which is not SE_2 -definable by Corollary 3.

- (b) The relations $Y = \{y = x^2\}$ and $Z = \{z = y^2\}$ in $\mathbb{Z}^3$ are defined by $y - x^2 = 0$ and $z - y^2 = 0$, without unknowns. Suppose $P(x, y, z, u)$ of total degree at most two, with r unknowns u , defines $Y \cap Z$. For integers x, z ,

$$z = x^4 \iff \exists y \exists u P(x, y, z, u) = 0 :$$

if $z = x^4$, then $(x, x^2, x^4) \in Y \cap Z$ and some u works with $y = x^2$; conversely $P(x, y, z, u) = 0$ gives $(x, y, z) \in Y \cap Z$, so $y = x^2$ and $z = y^2 = x^4$. Thus P , read with y as one more unknown, defines the graph of $x \mapsto x^4$ with $r + 1$ unknowns, contradicting Corollary 3. $\square$

In (b) the free variable y of the intersection is turned into an unknown; this is the closure under existential quantification noted above, so the failure lies in the conjunction. Corollary 4 shows that SE_2 lacks the closure under composition and conjunction on which iterated pairing normally relies; by itself it does not exclude an SE_2 coding of triples, which over $\mathbb{Z}$ is excluded by Theorem 1.

7. REMARKS

1. *What is not assumed.* Theorem 2 needs no nondegeneracy of Q , no bound on the number r of unknowns, no uniqueness of the unknowns, and no discriminant condition. Output uniqueness is used throughout: in §4.1 (to remove the unknowns, to exclude $\beta(a) = 0$ and to force $m \leq 1$), to find the centre (Lemma 2.2), to identify the section with f in §4.2, and as the contradiction in §4.5.

2. *Constructivity.* The proof of Theorem 2 is not constructive as written (it decides the branch over $\mathbb{Q}$ and chooses solutions); we have not tried to make it so.
3. *Inputs from outside.* Besides elementary algebra over $\mathbb{Q}$, the proof uses one external result: the existence of a non-trivial solution of the Pell equation (§4.5). The decidability results [Sie72, GS04] and the DPRM theorem are cited for context only and are not used.

APPENDIX A. CORRESPONDENCE WITH THE FORMAL DEVELOPMENT

Archived: in the development [Fuk-L], namespace `DiophCompression`. `ResearchUpgrade`, file given relative to `ResearchUpgrade/`. Supplementary: in [Fuk-S] (§1.4), file given relative to `SelectionS65/`, built against [Fuk-L] with the same toolchain and mathlib revision. “Same” means that the formal statement says what the paper says, up to notation; differences are noted. All declarations listed depend only on `propext`, `Classical.choice`, `Quot.sound`. Where marked †, a dependency check walked the transitive closure of the constants used in the declaration and confirmed that no constant name contains any string of a fixed list of excluded names: the decidability results for quadratic equations, the DPRM theorem, the projection theorem and the growth bounds of the archived development, and its conditional hypotheses. This is a check on names against that list, not a semantic test; the list and the eight checked declarations (the † rows) are given with the formal development.

Paper	Lean declaration	File	Statement vs paper
Theorem 1	<code>singleQuadratic_sharp_iff</code> †	FidelityBridge	same: $\exists$ injective SE_2 $f : \mathbb{Z}^n \rightarrow \mathbb{Z}^m \iff n \leq m$
Theorem 1 ($\Rightarrow$)	<code>singleQuadratic_injective_</code> <code>dimension_sharp</code> †	FidelityBridge	same
Theorem 2	<code>theorem2_sealed</code> †	Corr ([Fuk-S])	composition of the next two rows; linear growth with $C \in \mathbb{N}$
Theorem 2 (bridge)	<code>graphData_of_single</code> <code>QuadraticDefinable</code>	FidelityBridge	$\text{SE}_2 f \Rightarrow$ an equation of the shape (2.1) defining f (totality and output uniqueness)
Theorem 2 (body)	<code>summit_of_defines</code> †	Summit	defining equation $\Rightarrow$ alternative 1 or 2
Lemma 2.1	<code>range_polarMapQ_eq_dual</code> <code>Annihilator_ker</code>	Polar Annihilator	same, over $\mathbb{Q}$
Lemma 2.2	<code>zLinQ_radicalQ_eq_zero</code> , out <code>CoordQ_radicalQ_eq_zero</code>	RadicalBridge	same hypotheses; formal proof differs
Corollary 2.3	<code>exists_affine_centre_of_</code> <code>all_inputs</code> , <code>completeSquare</code>	AffineCentre, RationalCentre	same
Lemma 3.1	<code>dim_le_of_linearGrowth_</code> <code>injective</code>	LinearGrowth Dimension	same
Lemma 3.2	<code>not_injective_of_weakVector</code> <code>CosetNF</code>	DimensionFrom Coset	same
§4 branch split	<code>branch_exhaustive</code>	BranchSplit	same
§4.1	<code>weakVectorCosetNF_of_linear</code> <code>Branch_general</code>	Summit	same
§4.2 Step 1	<code>exists_isotropic_</code> <code>nonorthogonal_of_nonradical</code>	Isotropic Direction	same statement; construction differs

Paper	Lean declaration	File	Statement vs paper
§4.2 Steps 2–4	<code>exists_rational_solution_of_isotropic, weakVector CosetNF_of_isotropic</code>	RationalCentre, IsotropicCoset	same
§4.2(a), end of §4.5	<code>linearGrowth_of_offCentre_radical</code>	IsotropicBranch	same
§4.3, (4.1)	<code>linearGrowth_of_semidefinite_defines, out Coord_sq_le</code>	RadicalBridge, Output Coercivity	same; constant 4
§4.4	<code>linearGrowth_of_neg Semidfinite_defines</code>	Summit	same
§4.5	<code>exists_visible_orthogonal, not_isSquare_of_quotient Anisotropic, exists_pell_unit_congr †, pell_branch_contradiction</code>	PellPlane, Pell Nonsquare, Pell Step, Pell Branch	same; visible direction constructed differently; Pell existence taken from mathlib
§5	<code>exists_graph_injective_pad, atomGraphZZ_capacity_eq_one</code>	PadGraph, FidelityBridge	same; coding arity one
Corollary 3	<code>polyFn_SE2_iff †</code>	PolyGraph ([Fuk-S])	same, with $m = 1$, for p with integer coefficients; proof via Theorem 2 only. The extension to integer-valued p with rational coefficients (§6) is not formalized
Corollary 4(a)	<code>not_closed_under_composition †</code>	PolyGraph ([Fuk-S])	stated for functions $\mathbb{Z} \rightarrow \mathbb{Z}$
Corollary 4(b)	<code>not_closed_under_conjunction †</code>	Conjunction ([Fuk-S])	same relations in $\mathbb{Z}^3$

REFERENCES

- [Cor99] G. Cornelissen, Stockage diophantien et hypothèse abc généralisée, C. R. Acad. Sci. Paris Sér. I Math. 328 (1999), 3–8.
- [DPR61] M. Davis, H. Putnam, J. Robinson, The decision problem for exponential Diophantine equations, Ann. of Math. 74 (1961), 425–436.
- [FP23] R. Fueter, G. Pólya, Rationale Abzählung der Gitterpunkte, Vierteljschr. Naturforsch. Ges. Zürich 68 (1923), 380–386.
- [Fuk-L] H. Fukui, Structure and Computability of the Single-Quadratic Existential Fragment over the Integers (Lean 4 formalization), version 0.31.0, Zenodo, 2026, doi:10.5281/zenodo.23053094.
- [Fuk-S] H. Fukui, Single quadratic equations do not compress integer dimension: supplementary Lean 4 development, version 1.0.0, Zenodo, 2026, doi:10.5281/zenodo.23132400.
- [Gj21] K. S. Gjaldbæk, Non-injectivity of nonzero discriminant polynomials and applications to packing polynomials, arXiv:2102.03392 (2021).
- [GS04] F. Grunewald, D. Segal, On the integer solutions of quadratic equations, J. Reine Angew. Math. 569 (2004), 13–45.
- [Ham75] F. D. Hammer, Advanced problem 6028, Amer. Math. Monthly 82 (1975), no. 4, 410.
- [Lew81] J. S. Lew, Polynomials in two variables taking distinct integer values at lattice-points, Amer. Math. Monthly 88 (1981), 344–346.
- [LR78a] J. S. Lew, A. L. Rosenberg, Polynomial indexing of integer lattice-points I. General concepts and quadratic polynomials, J. Number Theory 10 (1978), 192–214.
- [Mat70] Yu. Matiyasevich, Enumerable sets are Diophantine, Dokl. Akad. Nauk SSSR 191 (1970), 279–282.
- [Nat16] M. B. Nathanson, Cantor polynomials and the Fueter–Pólya theorem, Amer. Math. Monthly 123 (2016), 1001–1012.
- [NZM91] I. Niven, H. S. Zuckerman, H. L. Montgomery, An Introduction to the Theory of Numbers, 5th ed., Wiley, New York, 1991.

- [Pas20] H. Pasten, Bivariate polynomial injections and elliptic curves, *Selecta Math. (N.S.)* 26 (2020), Paper No. 22.
- [Poo10] B. Poonen, Multivariable polynomial injections on rational numbers, *Acta Arith.* 145 (2010), 123–127.
- [Sie72] C. L. Siegel, Zur Theorie der quadratischen Formen, *Nachr. Akad. Wiss. Göttingen Math.-Phys. Kl. II* (1972), 21–46.
- [Sko38] Th. Skolem, *Diophantische Gleichungen*, Springer, Berlin, 1938.
- [Vse02] M. A. Vsemirnov, Two elementary proofs of the Fueter–Pólya theorem on pairing polynomials, *St. Petersburg Math. J.* 13 (2002), 705–715; Russian original *Algebra i Analiz* 13 (2001), no. 5, 1–15.

RESEARCH INSTITUTE OF CRIMINAL PSYCHIATRY / SEX OFFENDER MEDICAL CENTER,
TOKYO, JAPAN; AND DEPARTMENT OF NEUROPSYCHIATRY, KYOTO UNIVERSITY, KYOTO, JAPAN
Email address: `fukui@somec.org`